

EECS 16A Designing Information Devices and Systems I

Homework 10

Submission Format

Your homework submission should consist of **one** file.

- `hw10.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned)

Submit each file to its respective assignment on Gradescope.

1. Reading Assignment

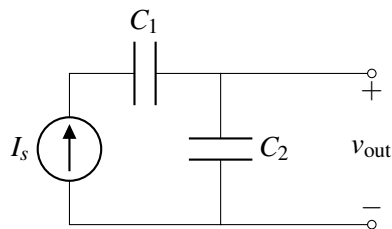
For this homework, please read Notes 18.1-18.6 and Notes 19.1-19.4, as well as Note 19.7. They will provide an overview of operational amplifiers (op-amps), negative feedback, the “golden rules” of op-amps, and some op-amp configurations (inverting and buffers). You are always encouraged to read beyond this as well.

- What are the two “golden rules” of ideal op amps? When do these rules hold true?
- What is the internal gain of an op-amp, A ? What is its value for an ideal op-amp? For non-ideal?

2. More Current Sources And Capacitors

(Contributors: Ava Tan, Panos Zarkos, Wahid Rahman)

- For the circuit given below, give an expression for $v_{\text{out}}(t)$ in terms of I_s , C_1 , C_2 , and time t . Assume that all capacitors are initially uncharged, i.e. the initial voltage across each capacitor is 0V.

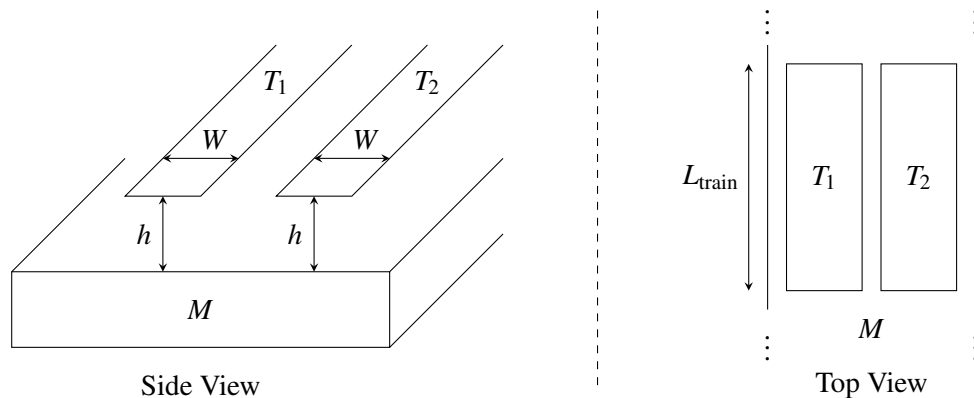


3. Maglev Train Height Control System

(Contributors: Laura Brink, Avi Pandey, Vijay Govindarajan, Titan Yuan, Lam Nguyen, Urmita Sikder)

One of the fastest forms of land transportation are trains that actually travel slightly elevated from the ground using magnetic levitation (or “maglev” for short). Ensuring that the train stays at a relatively constant height above its “tracks” (the tracks in this case are what provide the force to levitate the train and propel it forward) is critical to both the safety and fuel efficiency of the train. In this problem, we’ll explore how maglev trains could use capacitors to ensure correct elevation. (Note that real maglev trains may use completely different and much more sophisticated techniques to perform this function, so if you get a contract to build such a train, you’ll probably want to do more research on the subject.)

- (a) As shown below, we put two parallel strips of metal (T_1 , T_2) along the bottom of the train and we have one solid piece of metal (M) on the ground below the train (perhaps as part of the track).

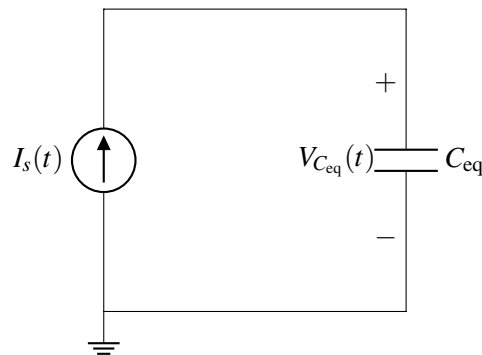


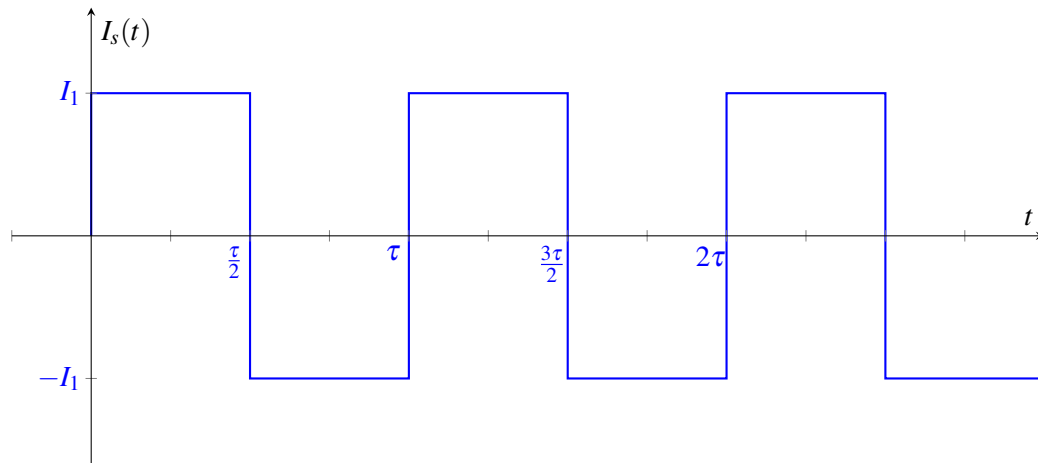
We assume that the entire train is at a uniform height above the track, so we can use the simple equations developed in lecture to model the capacitance.

As a function of L_{train} (the length of the train), W (the width of T_1 and T_2), and h (the height of the train above the track), determine the capacitances between T_1 and M and between T_2 and M . *Hint: Note that the area of a capacitor is given by the overlap area between its two plates.*

- (b) Any circuit on the train can only make direct contact at T_1 and T_2 . Thus, you can only measure the equivalent capacitance between T_1 and T_2 . Draw a circuit model showing how the capacitors between T_1 and M and between T_2 and M are connected to each other.
- (c) Using the same parameters as in part (a), provide an expression for the equivalent capacitance measured between T_1 and T_2 .
- (d) We want to build a circuit that creates a voltage waveform with an amplitude that changes based on the height of the train. Your colleague recommends you start with the circuit as shown below, where I_s is a periodic current source, and C_{eq} is the equivalent capacitance between T_1 and T_2 . The graph below shows $I_s(t)$, a square wave with period τ and amplitude I_1 , as a function of time t .

Find an equation for and draw the voltage $V_{C_{\text{eq}}}(t)$ as a function of time t . Assume the capacitor C_{eq} is discharged at time $t = 0$, so $V_{C_{\text{eq}}}(0) = 0 \text{ V}$.



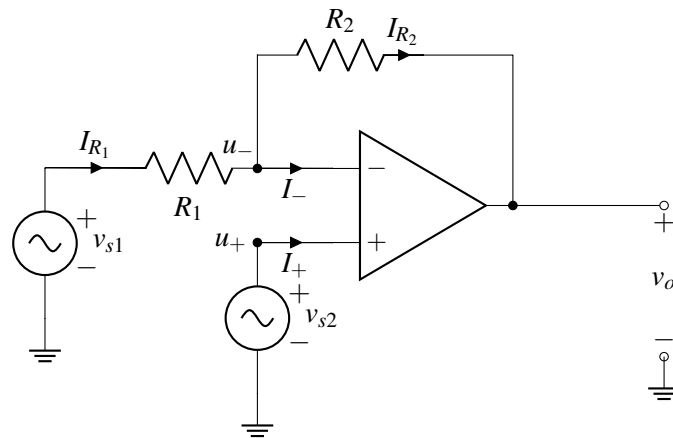


- (e) We now want to develop an indicator that alerts us when the train is too high above the tracks. We want to have an output of 5 V (to trigger the alert) when the height of the train h is above 1 cm, and an output of 0 V when h is below 1 cm.

We will assume the train has length $L_{\text{train}} = 100\text{ m}$ and that the metals, T_1 and T_2 , have width $W = 1\text{ cm}$, and the space between the train and the solid piece of metal M has permittivity $\epsilon = 8.85 \times 10^{-12}\text{ F m}^{-1}$. Design a circuit using **a square wave current source (i.e. I_s in part (d)) with period $\tau = 1\text{ }\mu\text{s}$ and pulses of amplitude $I_1 = 1\text{ mA}$, a comparator, and any number of voltage sources** to implement this function. *Hint: You should use the circuit you analyzed in part (d).*

4. Amplifier with Multiple Inputs

Consider the circuit below, assuming the op-amp is ideal.



- Is this circuit in negative feedback? Why or why not?
- What is the current I_- flowing into the negative terminal of the op-amp?
- What is the current I_+ flowing into the positive terminal of the op-amp?
- What is u_- , the voltage at the negative terminal of the op-amp, in terms of v_o and v_{s1} ?
- Prove the second golden rule, that $u_+ = u_-$, assuming the op-amp in the circuit above is ideal.

Hint: Recall that $v_o = A(u_+ - u_-)$, where A is the gain of the op-amp.

Hint: Try replacing $\left(\frac{R_2}{R_1 + R_2}\right)$ with a constant f .

- (f) What is u_- in terms of v_{s2} ?
- (g) What is I_{R_2} , the current through the resistor R_2 , in terms of v_{s1} , v_{s2} , R_1 , and R_2 ? *Hint:* Use KCL at node u_- .

5. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.